

PROJECT REPORT

(INT375: DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING)

On

YouTube Content Data Analysis

Submitted by:

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In partial fulfilment for the requirement of the award of the degree
of
“B.Tech. (Computer Science and Technology)”



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Submitted To Dr. Mrinalini Rana,

Lovely Professional University
Jalandhar, Punjab, India

CERTIFICATE

This is to certify that **Deepak Rao** bearing Registration no.**12401110** has completed **INT375** project titled “**YouTube Content Data Analysis**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

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Date: 22th of April 2026

DECLARATION

I, **Deepak Rao** student of **P132: B.Tech (Computer Science and Engineering)** under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

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Through this course, I gained knowledge in important areas such as data cleaning, data preprocessing, data visualization, hypothesis testing, and regression analysis. These concepts were essential in completing this project successfully and in developing a deeper understanding of how data science can be applied to real-world problems.

I would also like to thank my institution for providing the necessary resources, infrastructure, and learning environment that supported me in carrying out this project. The availability of tools and platforms enabled me to implement practical solutions and enhance my technical skills.

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Correlation Analysis of Video Engagement Metrics

	Views	Likes	Comments	Subscribers
Views	1.00	0.85	0.17	0.50
Likes	0.85	1.00	0.31	0.61
Comments	0.17	0.31	1.00	0.16
Subscribers	0.50	0.61	0.16	1.00

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I. INTRODUCTION

In today's digital age, online platforms have become a major part of how people create, share, and consume content. Among these platforms, YouTube stands out as one of the most popular spaces for video sharing and communication. Every day, millions of videos are uploaded, making it important to understand how audiences interact with this content.

User engagement on YouTube is usually measured using simple indicators such as views, likes, comments, and subscribers. These metrics give a clear idea of how people respond to a video. For example, views show how many people have watched a video, while likes and comments reflect how much the audience is interested or involved. Looking at these metrics together helps in understanding how well a video performs.

With such a large amount of data available, it becomes difficult to analyse everything manually. This is where data science becomes useful. It provides tools and techniques that help in organising, analysing, and interpreting data in a meaningful way. In this project, methods such as data preprocessing, exploratory data analysis (EDA), visualisation, hypothesis testing, and linear regression have been used to study engagement patterns.

The main aim of this project is to understand what factors influence engagement on YouTube videos. By analysing the relationship between views, likes, comments, and subscriber count, the study attempts to identify patterns that can explain why some videos perform better than others. Statistical methods are also used to check whether these patterns are reliable and not just due to chance.

Overall, this project focuses on applying simple data analysis techniques to a real-world dataset. It shows how data can be used to understand user behaviour and make better decisions when creating or managing content on digital platforms.

II. LITERATURE REVIEW

Over the past decade, the growth of digital platforms has transformed the way information is created, shared, and consumed. Among these platforms, YouTube has emerged as one of the most influential mediums for content distribution. As a result, analysing user engagement has become an important area of study, particularly for understanding how audiences respond to different types of content.

User engagement on YouTube is commonly measured through indicators such as views, likes, comments, and subscriber count. These metrics provide a quantitative representation of audience behaviour and are widely used to evaluate the performance of video content. Previous studies suggest that while views indicate the reach of a video, likes and comments reflect a deeper level of audience interaction and satisfaction. Therefore, analysing these metrics together offers a more complete understanding of engagement.

A number of researchers have examined the relationship between these engagement indicators. It has been observed that videos with higher view counts often receive more likes and comments, indicating a positive relationship between visibility and interaction. However, this relationship is not always linear, as other factors such as content quality, relevance, and audience preferences also play a significant role. This highlights the need for data-driven approaches to better understand these patterns.

In addition to engagement metrics, content-related factors have also been identified as important contributors to video performance. Studies indicate that elements such as content type, presentation style, and timing of uploads can significantly influence audience behaviour. For instance, videos that are informative, visually appealing, or aligned with current trends tend to attract more interaction. At the same time, audience behaviour varies across individuals, with some users actively engaging through likes and comments, while others remain passive viewers.

To analyse these complex relationships, data science techniques are widely applied in recent research. Exploratory Data Analysis (EDA) is often used as an initial step to understand the structure and characteristics of the dataset. Visualisation techniques such as bar charts, scatter plots, and line graphs help in identifying trends and patterns in the data. These visual tools make it easier to interpret large datasets and communicate findings effectively.

At this stage, you may include your own visualisations to support the discussion. For example, a correlation heatmap can be used to illustrate the relationship between key variables such as views, likes, and comments.

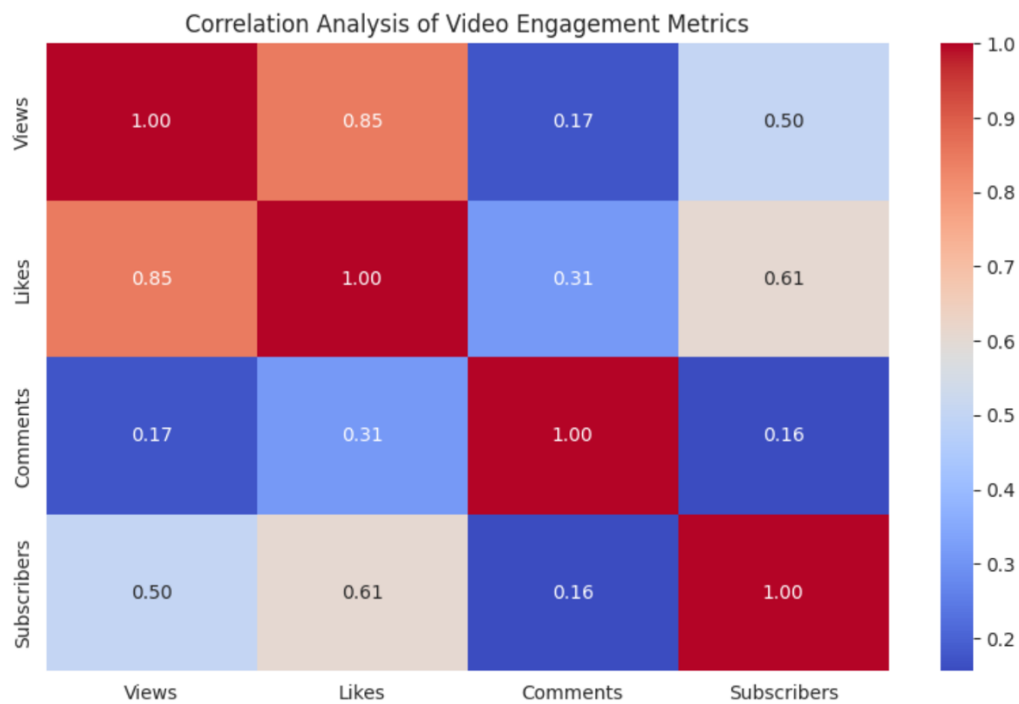


Fig. 1. Correlation Analysis of Video Engagement Metrics

Such visual representations often show a strong positive association between views and likes, suggesting that increased visibility leads to higher engagement. Similarly, scatter plots can be used to examine the relationship between views and likes in more detail.

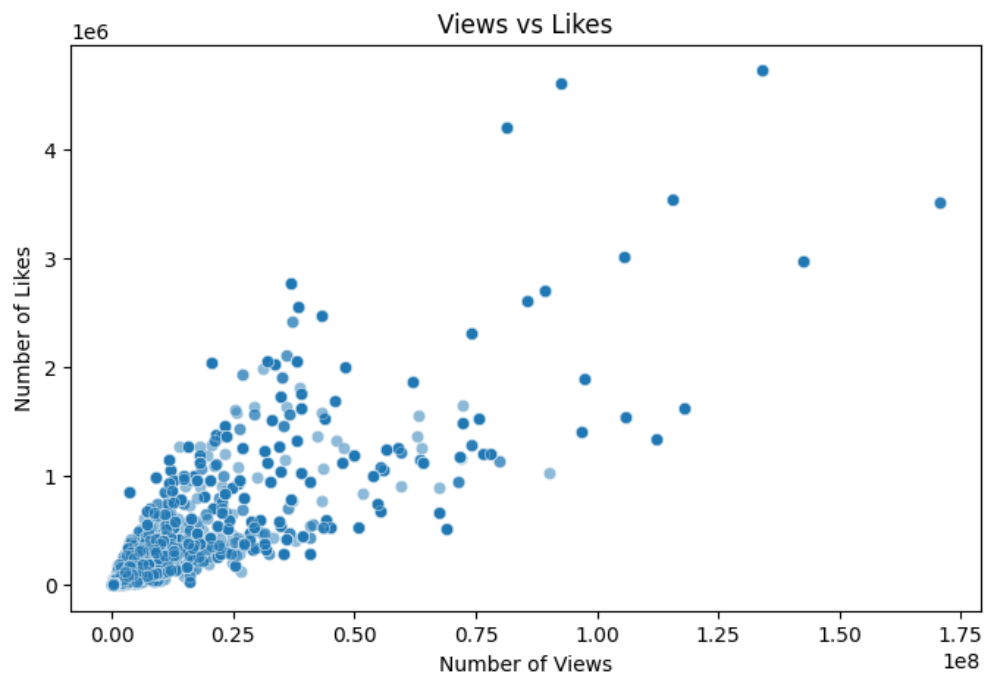


Fig. 2. Relationship Between Views and Likes

Beyond visual analysis, statistical methods are also used to validate patterns observed in the data. Hypothesis testing, such as the t-test, helps determine whether differences between groups are statistically significant. Regression analysis, on the other hand, is used to model the relationship between variables and predict outcomes. These techniques provide a more structured and reliable approach to analysing engagement data.

Overall, the existing literature indicates that user engagement on YouTube is influenced by a combination of factors, including visibility, interaction, and content characteristics. While previous studies have explored these aspects in isolation, there remains a need to apply practical data analysis techniques to real-world datasets. This project builds on these ideas by using data science methods to examine engagement patterns and draw meaningful conclusions.

III. OBJECTIVES

The main objectives of this project are as follows:

1. To analyse the relationship between key engagement metrics such as views, likes, comments, and subscriber count on YouTube videos.
2. To understand how different factors influence user engagement and video performance.
3. To identify patterns and trends in the data using exploratory data analysis and visualisation techniques.
4. To compare engagement levels across different video categories using statistical methods.
5. To apply hypothesis testing in order to determine whether differences in engagement are statistically significant.
6. To build a simple linear regression model to examine how one variable, such as views, affects another, such as likes.
7. To interpret the results in a clear and meaningful way so that they can be used to understand audience behaviour and improve content strategies.

V. DATASET DESCRIPTION

The dataset used in this project is a large-scale YouTube trending videos dataset obtained from Kaggle. It contains detailed information about video performance across multiple regions and provides a wide range of features related to user engagement and channel statistics. The dataset is designed to capture how videos perform on a global scale, making it suitable for analysing patterns in audience behaviour and content popularity.

One of the most important aspects of this dataset is its size. The complete dataset is approximately **17.65 GB**, which classifies it as a large dataset. Due to its size, it cannot be easily handled using standard local computing systems with limited memory and processing power. Attempting to load the entire dataset at once may lead to performance issues or system crashes. Therefore, it was necessary to use an alternative approach for handling and analysing the data.

[Kaggle Dataset \[1\]](#)

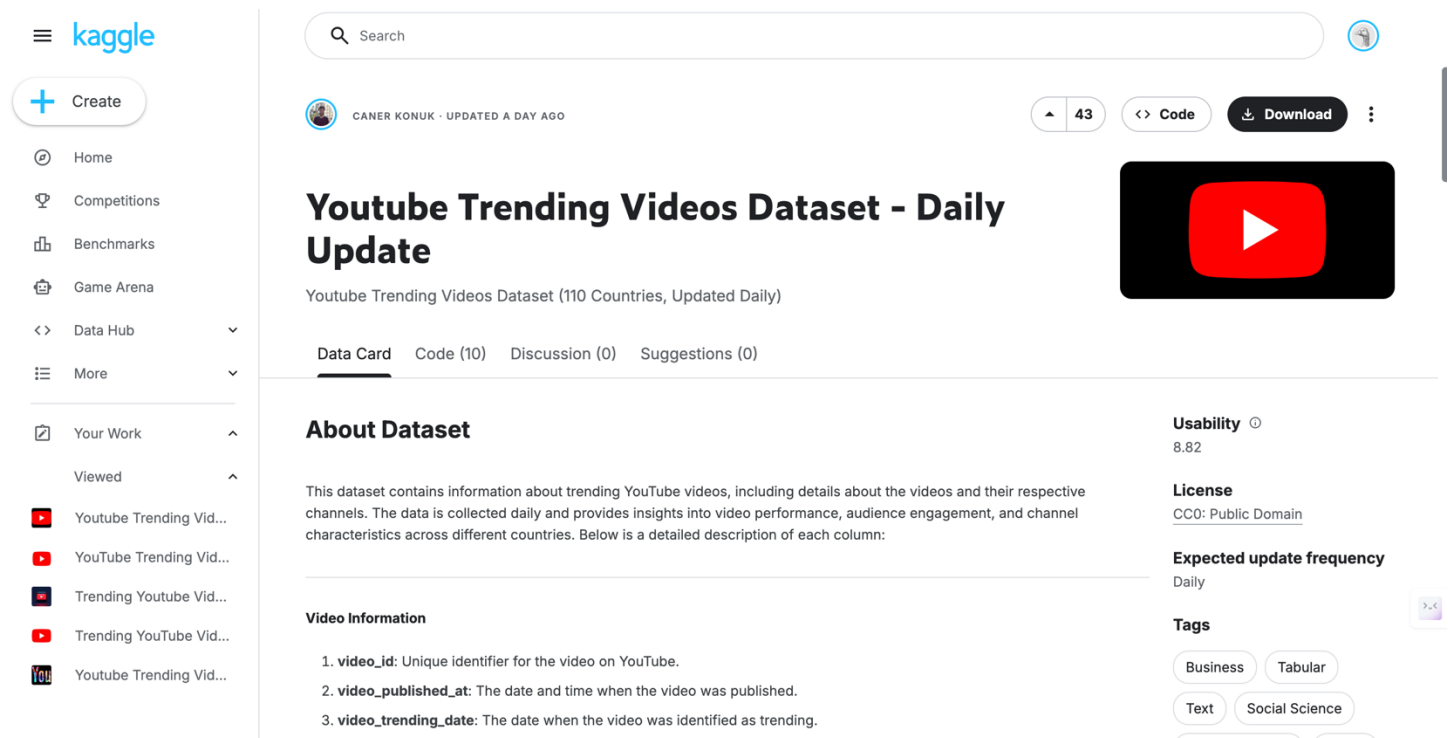
The image is a screenshot of the Kaggle website showing a dataset page. On the left is a sidebar with the Kaggle logo and navigation links: Home, Competitions, Benchmarks, Game Arena, Data Hub, More, Your Work, and Viewed. The main content area has a search bar at the top. Below it, the dataset title 'Youtube Trending Videos Dataset - Daily Update' is displayed, along with the uploader's name 'CANER KONUK' and the update status 'UPDATED A DAY AGO'. There are buttons for '43' (likes), 'Code', and 'Download'. A large red play button icon is visible. Below the title, it says 'Youtube Trending Videos Dataset (110 Countries, Updated Daily)'. There are tabs for 'Data Card', 'Code (10)', 'Discussion (0)', and 'Suggestions (0)'. The 'About Dataset' section describes the data: 'This dataset contains information about trending YouTube videos, including details about the videos and their respective channels. The data is collected daily and provides insights into video performance, audience engagement, and channel characteristics across different countries. Below is a detailed description of each column:'. Under 'Video Information', there are three items: 1. video_id: Unique identifier for the video on YouTube. 2. video_published_at: The date and time when the video was published. 3. video_trending_date: The date when the video was identified as trending. On the right side, there are sections for 'Usability' (8.82), 'License' (CC0: Public Domain), 'Expected update frequency' (Daily), and 'Tags' (Business, Tabular, Text, Social Science).

Fig. 3. Dataset Source from Kaggle

To address this challenge, **Google Colab** was used as the primary working environment. Google Colab provides a cloud-based platform that supports Python programming and allows users to execute code without relying on local system resources. It is especially useful for handling large datasets, as it provides access to higher memory and processing capabilities compared to typical personal computers.

In addition to Google Colab, **Google Drive** was used for storing and managing the dataset. Since the dataset size is quite large, storing it locally would not be practical. By uploading the dataset to Google Drive, it became easier to access the file directly from Colab. This setup allowed seamless integration between storage and processing, making the workflow more efficient and organised.

Dataset

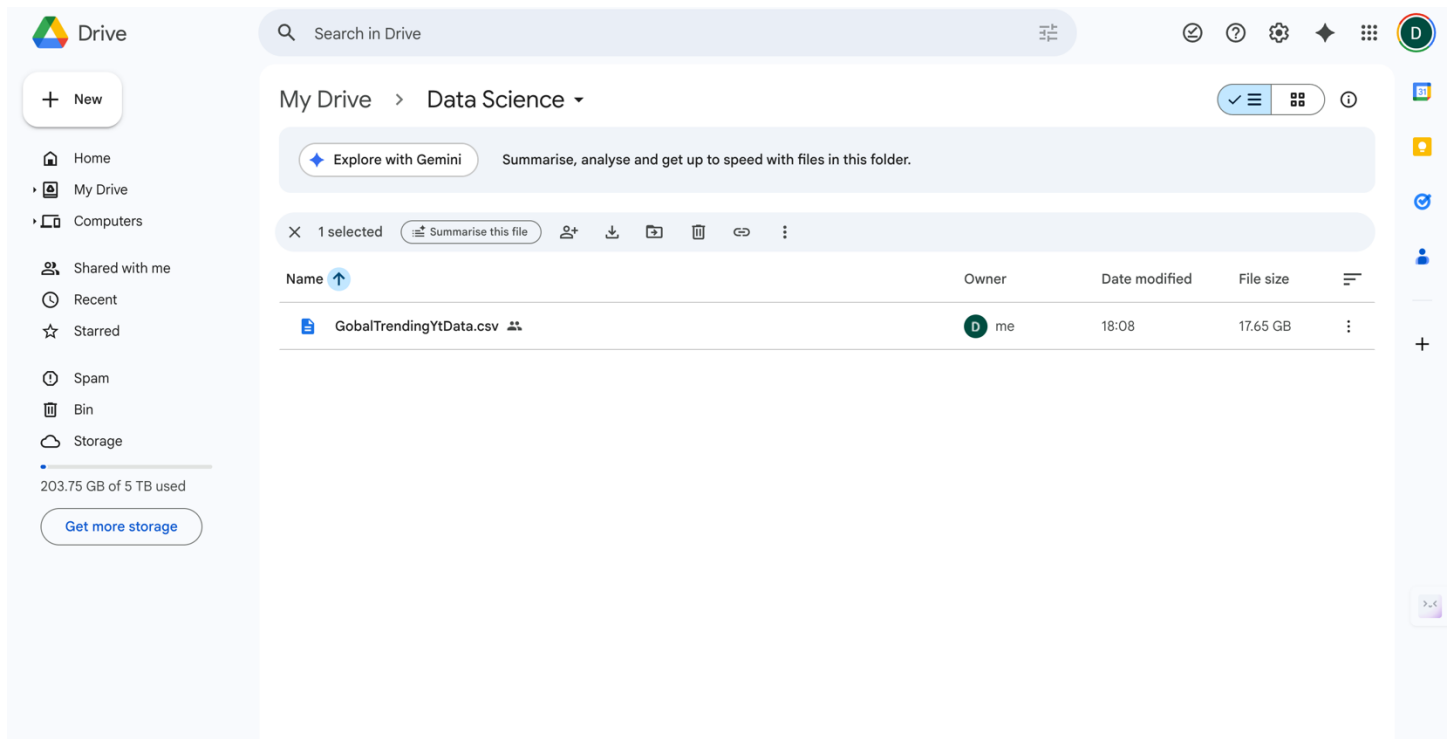


Fig. 4. Dataset Stored in Google Drive

The dataset was accessed in Google Colab using the following code:

```
from google.colab import drive
drive.mount('/content/drive')
```

Code Snippet 1. Mounting Google Drive and loading dataset

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
```

```
# Change this path if your file is inside a specific folder in your Drive
file_path = '/content/drive/MyDrive/Data Science/GobalTrendingYtData.csv'
```

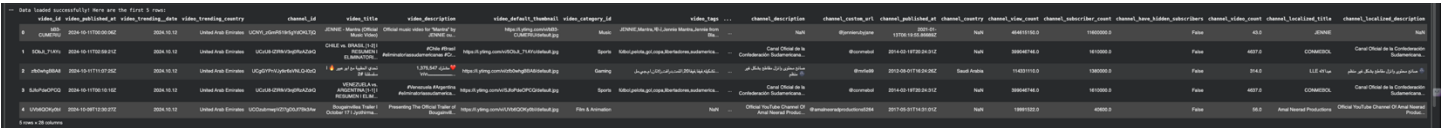
```
# Load only the first 10,000 rows
df = pd.read_csv(file_path, nrows=10000)
```

Due to memory limitations even within cloud environments, it was not feasible to work with the entire dataset at once. Therefore, only a subset of the dataset was used for analysis. A limited number of rows were loaded using techniques such as specifying the number of rows while reading the file. This approach ensured that the analysis could be performed smoothly without exceeding memory limits, while still maintaining a representative sample of the data.

```
print("Data loaded successfully! Here are the first 5 rows:")
df.head()
```

Code Snippet 2. Loading subset of dataset using nnows

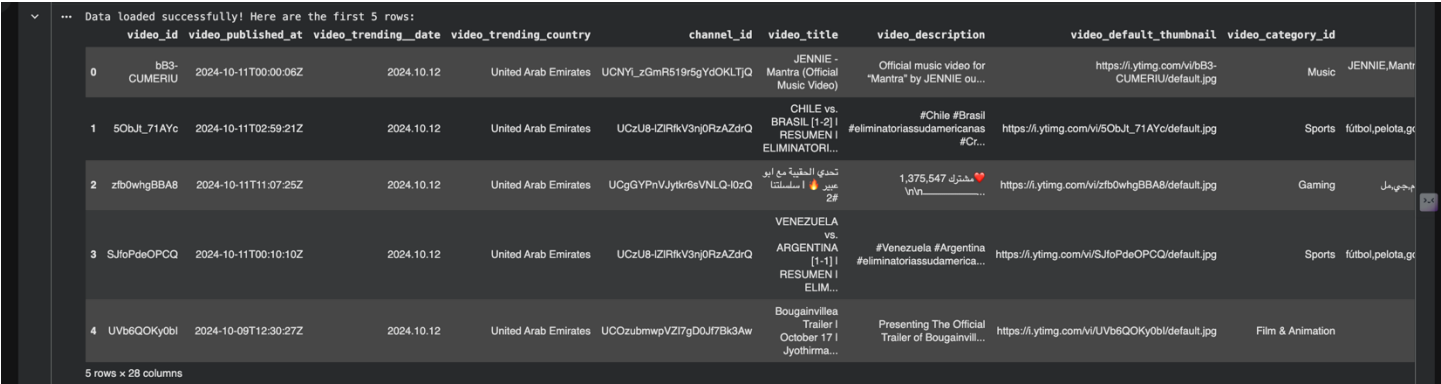
The dataset contains a variety of features that describe both the video and the channel associated with it. These features are essential for understanding how users interact with content and how engagement is distributed across different videos.



	video_id	video_published_at	video_trending_date	video_trending_country	channel_id	video_title	video_description	video_default_thumbnail	video_category_id	video_views	video_likes	video_comments	channel_subscriber_count	channel_video_count	channel_total_views	channel_total_likes
0	hB3-CUMERIU	2024-10-11T00:00:06Z	2024.10.12	United Arab Emirates	UCNYLzGmR519r5gYdOKLTJQ	JENNIE - Mantra (Official Music Video)	Official music video for "Mantra" by JENNIE ou...	https://i.ytimg.com/vi/hB3-CUMERIU/default.jpg	Music	1,375,547	13,755	1,375	1,375,547	13,755	1,375,547	13,755
1	50bJL_71AYc	2024-10-11T02:59:21Z	2024.10.12	United Arab Emirates	UCzU8-IzIRkV3nj0RzAZdRQ	CHILE vs. BRASIL [1-2] RESUMEN ELIMINATORIA...	#Chile #Brasil #eliminatoriasudamericanas #Cr...	https://i.ytimg.com/vi/50bJL_71AYc/default.jpg	Sports	1,375,547	13,755	1,375	1,375,547	13,755	1,375,547	13,755
2	zfb0whgBBA8	2024-10-11T11:07:25Z	2024.10.12	United Arab Emirates	UCgGYPhVJykr6sVNLQ-i0zQ	تحتوي الحقيبة مع أبو عيسى (مسلسلتي) 2#	1,375,547 مشتركين	https://i.ytimg.com/vi/zfb0whgBBA8/default.jpg	Gaming	1,375,547	13,755	1,375	1,375,547	13,755	1,375,547	13,755
3	SJfoPdeOPCQ	2024-10-11T00:10:10Z	2024.10.12	United Arab Emirates	UCzU8-IzIRkV3nj0RzAZdRQ	VENEZUELA vs. ARGENTINA [1-1] RESUMEN ELIM...	#Venezuela #Argentina #eliminatoriasudamerica...	https://i.ytimg.com/vi/SJfoPdeOPCQ/default.jpg	Sports	1,375,547	13,755	1,375	1,375,547	13,755	1,375,547	13,755
4	UVb6QOKyObI	2024-10-09T12:30:27Z	2024.10.12	United Arab Emirates	UCOzubmwpVZ77gD0J7Bk3Aw	Bougainvillea Trailer October 17 Jyothirma...	Presenting The Official Trailer of Bougainvillea...	https://i.ytimg.com/vi/UVb6QOKyObI/default.jpg	Film & Animation	1,375,547	13,755	1,375	1,375,547	13,755	1,375,547	13,755

Fig. 5. Preview of Dataset (First 5 Rows Output)

Zoomed



	video_id	video_published_at	video_trending_date	video_trending_country	channel_id	video_title	video_description	video_default_thumbnail	video_category_id
0	hB3-CUMERIU	2024-10-11T00:00:06Z	2024.10.12	United Arab Emirates	UCNYLzGmR519r5gYdOKLTJQ	JENNIE - Mantra (Official Music Video)	Official music video for "Mantra" by JENNIE ou...	https://i.ytimg.com/vi/hB3-CUMERIU/default.jpg	Music
1	50bJL_71AYc	2024-10-11T02:59:21Z	2024.10.12	United Arab Emirates	UCzU8-IzIRkV3nj0RzAZdRQ	CHILE vs. BRASIL [1-2] RESUMEN ELIMINATORIA...	#Chile #Brasil #eliminatoriasudamericanas #Cr...	https://i.ytimg.com/vi/50bJL_71AYc/default.jpg	Sports
2	zfb0whgBBA8	2024-10-11T11:07:25Z	2024.10.12	United Arab Emirates	UCgGYPhVJykr6sVNLQ-i0zQ	تحتوي الحقيبة مع أبو عيسى (مسلسلتي) 2#	1,375,547 مشتركين	https://i.ytimg.com/vi/zfb0whgBBA8/default.jpg	Gaming
3	SJfoPdeOPCQ	2024-10-11T00:10:10Z	2024.10.12	United Arab Emirates	UCzU8-IzIRkV3nj0RzAZdRQ	VENEZUELA vs. ARGENTINA [1-1] RESUMEN ELIM...	#Venezuela #Argentina #eliminatoriasudamerica...	https://i.ytimg.com/vi/SJfoPdeOPCQ/default.jpg	Sports
4	UVb6QOKyObI	2024-10-09T12:30:27Z	2024.10.12	United Arab Emirates	UCOzubmwpVZ77gD0J7Bk3Aw	Bougainvillea Trailer October 17 Jyothirma...	Presenting The Official Trailer of Bougainvillea...	https://i.ytimg.com/vi/UVb6QOKyObI/default.jpg	Film & Animation

The **video view count** represents the total number of times a video has been watched. This metric is often used as an indicator of a video's reach or popularity. A higher number of views suggests that the video has been exposed to a larger audience.

The **video like count** indicates how many users have liked a video. This metric reflects the level of appreciation or positive feedback from viewers.

The **video comment count** shows the number of comments posted on a video. Comments represent a deeper level of interaction, as they require users to actively engage with the content.

The **channel subscriber count** represents the total number of subscribers of the channel that uploaded the video. This metric provides an indication of the channel's popularity and long-term audience base.

The **video category ID** classifies videos into different categories such as entertainment, education, and music. This helps in comparing performance across different types of content.

Table X: Description of Dataset Features

Video-Level Features

- **video_description**
This column contains the textual description of the video provided by the creator. It often includes additional information about the content, links, or context related to the video.
- **video_default_thumbnail**
This represents the default thumbnail image of the video. It is usually a URL pointing to the image shown as a preview before the video is played.
- **video_category_id**
This indicates the category to which the video belongs, such as entertainment, music, education, or news. It helps in grouping and comparing different types of content.
- **video_tags**
This field contains keywords or tags associated with the video. These tags help in improving searchability and are used by the platform to recommend content.

Channel-Level Features

- **channel_description**
This column contains the description of the YouTube channel. It provides information about the type of content the channel produces and its purpose.
- **channel_custom_url**
This is the custom URL of the channel, which is often personalised and easier to remember compared to the default channel link.
- **channel_published_at**
This indicates the date and time when the channel was created. It helps in understanding how long the channel has been active.
- **channel_country**
This represents the country associated with the channel. It can be useful for analysing regional trends and audience distribution.
- **channel_view_count**
This shows the total number of views received by all videos on the channel. It reflects the overall reach and popularity of the channel.
- **channel_subscriber_count**
This indicates the number of users who have subscribed to the channel. It is a key metric for measuring the channel's audience base.
- **channel_have_hidden_subscribers**
This is a boolean value (True/False) that indicates whether the channel has chosen to hide its subscriber count from public view.
- **channel_video_count**
This represents the total number of videos uploaded by the channel. It gives an idea of how active the channel is.
- **channel_localized_title**
This contains the channel's title in a localised or region-specific format, making it easier for users in different regions to understand.
- **channel_localized_description**
This provides the channel description in a localised format, similar to the title, to improve accessibility for different audiences.

Before performing any analysis, it is important to understand the structure and quality of the dataset. Initial inspection was carried out to examine the number of rows and columns, data types, and presence of missing values.

```
print("\n--- Data Information ---")
print(df.info())

print("\n--- Missing Values ---")
print(df.isnull().sum())

print("\n--- Quick Math Summary ---")
print(df.describe())
```

Code Snippet 3. Basic dataset inspection (info, describe, columns)

```
--- Data Information ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 28 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   video_id                             10000 non-null  object
 1   video_published_at                   10000 non-null  object
 2   video_trending_date                  10000 non-null  object
 3   video_trending_country               10000 non-null  object
 4   channel_id                           10000 non-null  object
 5   video_title                          10000 non-null  object
 6   video_description                    8811 non-null   object
 7   video_default_thumbnail              10000 non-null  object
 8   video_category_id                   9994 non-null   object
 9   video_tags                           7320 non-null   object
10   video_duration                       10000 non-null  object
11   video_dimension                      10000 non-null  object
12   video_definition                     10000 non-null  object
13   video_licensed_content               10000 non-null  bool
14   video_view_count                     10000 non-null  float64
15   video_like_count                     9827 non-null   float64
16   video_comment_count                  9940 non-null   float64
17   channel_title                        10000 non-null  object
18   channel_description                  9243 non-null   object
19   channel_custom_url                   9991 non-null   object
20   channel_published_at                 10000 non-null  object
21   channel_country                      8417 non-null   object
22   channel_view_count                   10000 non-null  float64
23   channel_subscriber_count             10000 non-null  float64
24   channel_have_hidden_subscribers      10000 non-null  bool
25   channel_video_count                  10000 non-null  float64
26   channel_localized_title              10000 non-null  object
27   channel_localized_description         9243 non-null   object
dtypes: bool(2), float64(6), object(20)
memory usage: 2.0+ MB
None
```

Fig. 6. Dataset Information Output

Since the dataset was collected from a real-world platform, it contains certain imperfections such as missing values and variations in data formatting. Handling these issues is an essential step to ensure that the analysis is accurate and reliable.

The dataset provides a rich source of information for studying user engagement on YouTube. Its large size and diverse features make it suitable for identifying patterns and relationships between different variables. However, working with such a dataset also requires careful handling and efficient use of tools to avoid computational limitations.

Overall, this dataset forms the foundation of the project and enables a detailed analysis of video engagement trends. By using cloud-based tools and appropriate data handling techniques, it becomes possible to extract meaningful insights from a large and complex dataset.

A. Correlation Heatmap

Graph

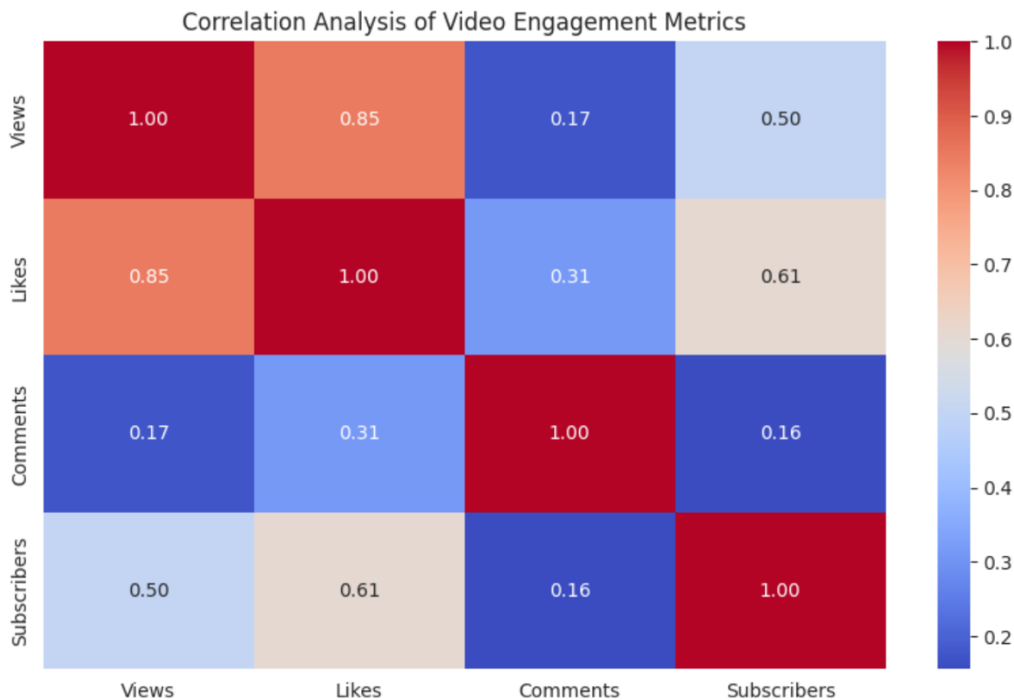


Fig. X. Correlation Analysis of Video Engagement Metrics

Interpretation

The correlation heatmap was used to understand the relationship between key numerical features in the dataset, namely views, likes, comments, and subscriber count. It provides a visual representation of how strongly these variables are related to each other.

From the heatmap, it can be observed that there is a **strong positive correlation between video view count and video like count (approximately 0.85)**. This indicates that videos with a higher number of views are more likely to receive a greater number of likes. This suggests that visibility plays an important role in user engagement.

The relationship between **likes and subscriber count is moderately positive (around 0.61)**. This shows that channels with more subscribers tend to receive more likes, although the effect is not as strong as views. This implies that while subscriber base contributes to engagement, other factors such as content quality and relevance also influence audience response.

On the other hand, the correlation between **comments and other variables is relatively weak**. For example, the relationship between views and comments is low, indicating that not all viewers actively participate by commenting. This highlights the difference between passive engagement (views) and active engagement (comments).

Overall, the heatmap shows that **views are the most influential factor affecting likes**, while comments and subscriber count have a comparatively smaller impact. These insights support further analysis using visualisation and regression techniques.

VIII. DATA VISUALISATION

A. Average Likes by Video Category

Graph

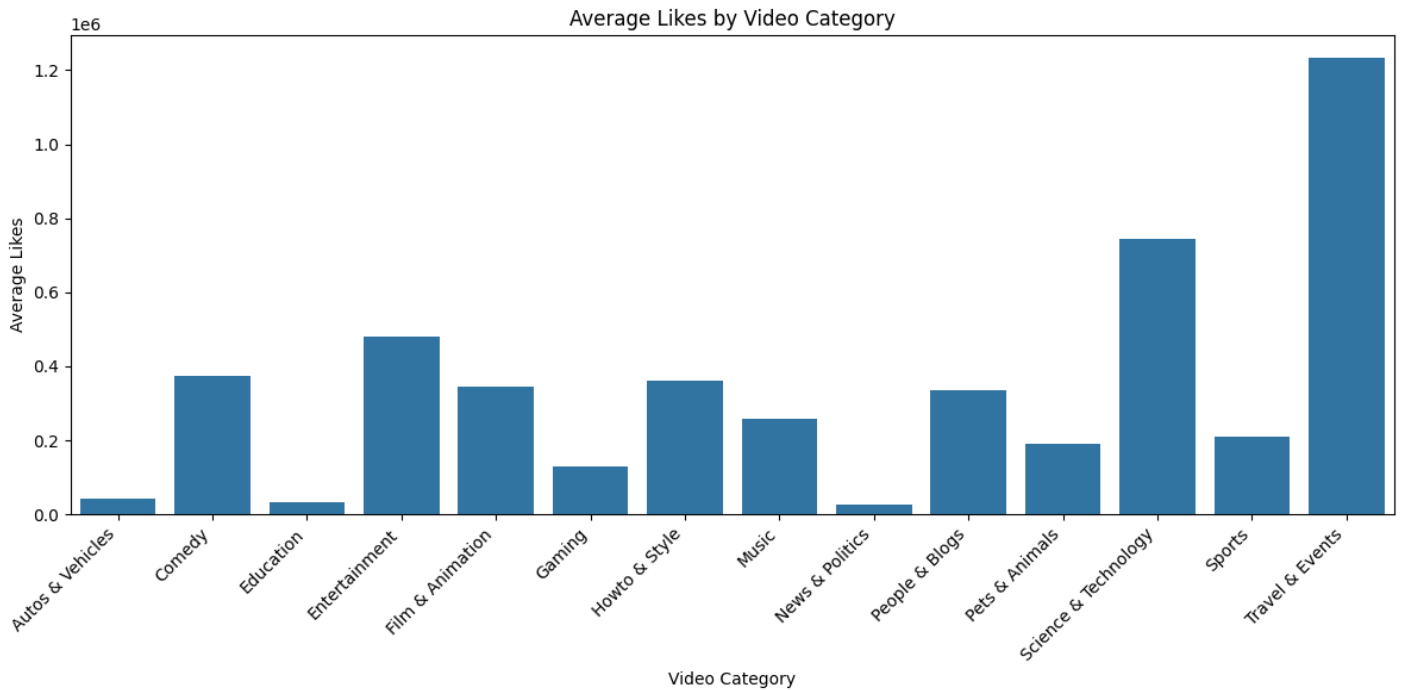


Fig. 8. Average Likes by Video Category

Code

```
plt.figure(figsize=(12,6)) # increase size
sns.barplot(data=avg_likes, x='video_category_id', y='video_like_count')

plt.title('Average Likes by Video Category')
plt.xlabel('Video Category')
plt.ylabel('Average Likes')

plt.xticks(rotation=45, ha='right') # rotate labels
plt.tight_layout() # auto adjust spacing

plt.show()
```

Interpretation

This graph shows the average number of likes received by videos across different categories. It helps in identifying which types of content attract more engagement from users.

It can be observed that categories such as **Travel & Events** and **Science & Technology** receive higher average likes compared to others. This suggests that users are more interested in content that is either informative or

experience-based. On the other hand, categories like **Education** and **News & Politics** show lower engagement, possibly due to a more specific audience.

Overall, the graph indicates that **content category** plays an important role in determining user engagement.

B. Views vs Likes

Graph

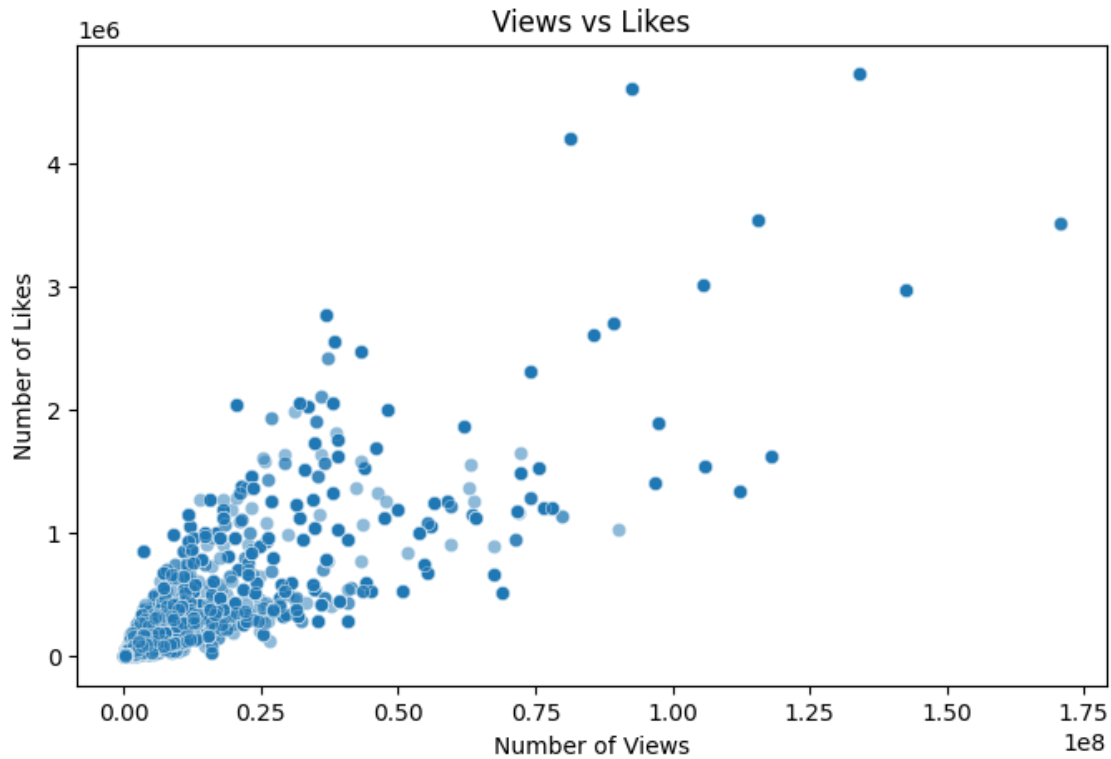


Fig. 9. Relationship Between Views and Likes

Code

```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='video_view_count', y='video_like_count',
alpha=0.5)
plt.title('Views vs Likes')
plt.xlabel('Number of Views')
plt.ylabel('Number of Likes')
plt.show()
```

Interpretation

The scatter plot shows the relationship between the number of views and likes for each video.

A clear **positive trend** can be seen, indicating that videos with more views generally receive more likes. However, the points are scattered, which means the relationship is not perfectly linear. This suggests that other factors such as content quality and audience interest also influence likes.

Overall, this graph confirms that **views are a strong factor influencing engagement**.

C. Average Likes vs Number of Comments

Graph

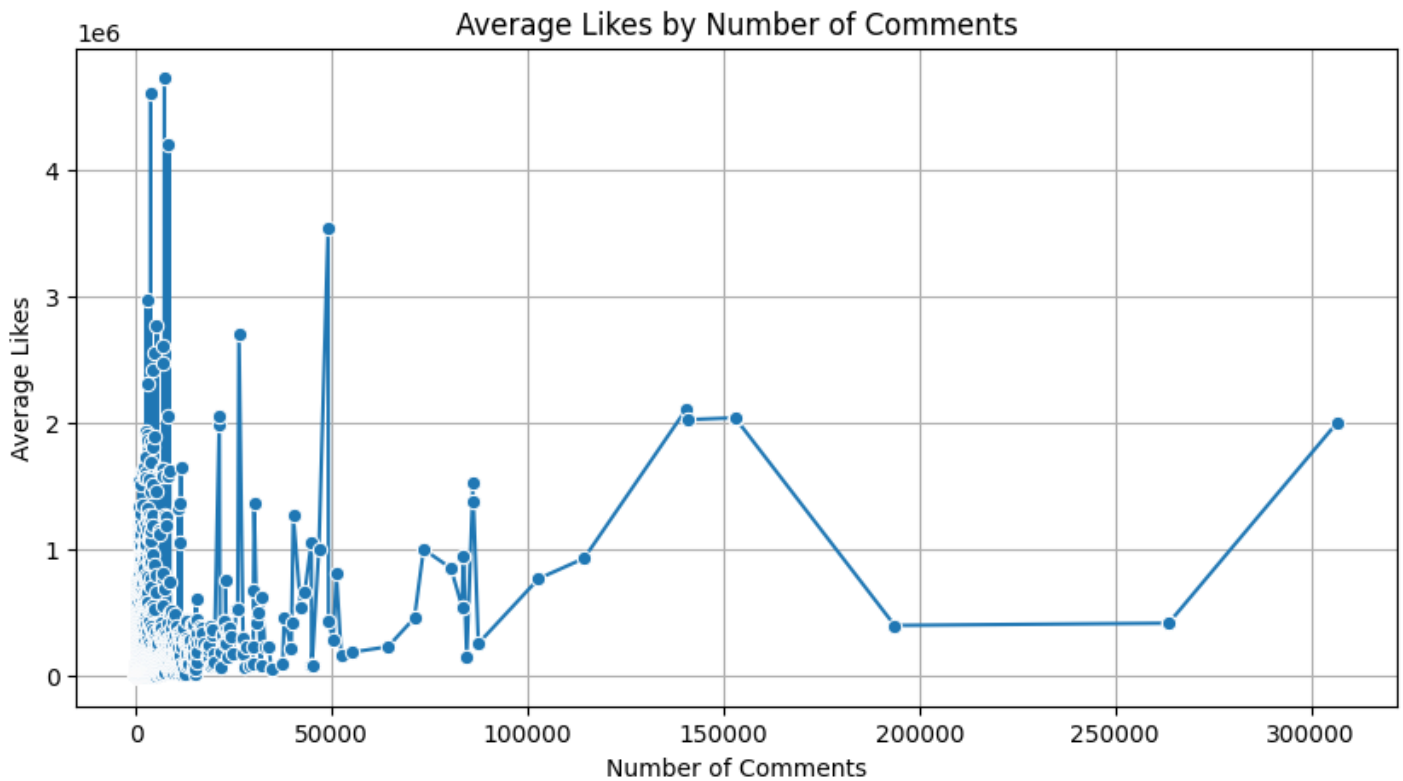


Fig. 10. Average Likes by Number of Comments

Code

```
avg_comment_likes =  
df.groupby('video_comment_count')['video_like_count'].mean().reset_index()  
  
plt.figure(figsize=(10, 5))  
sns.lineplot(data=avg_comment_likes, x='video_comment_count',  
y='video_like_count', marker='o')  
plt.title('Average Likes by Number of Comments')  
plt.xlabel('Number of Comments')  
plt.ylabel('Average Likes')  
plt.grid(True)  
plt.show()
```

Interpretation

This graph examines how the number of comments relates to the number of likes.

It can be seen that there is some increase in likes as comments increase, but the pattern is not consistent. Many points are scattered, especially at lower comment values. This indicates that while comments represent user interaction, they do not always lead to higher likes.

Overall, the relationship between comments and likes is **weaker compared to views and likes**.

D. Top 10 Countries: Content Category Performance

Graph

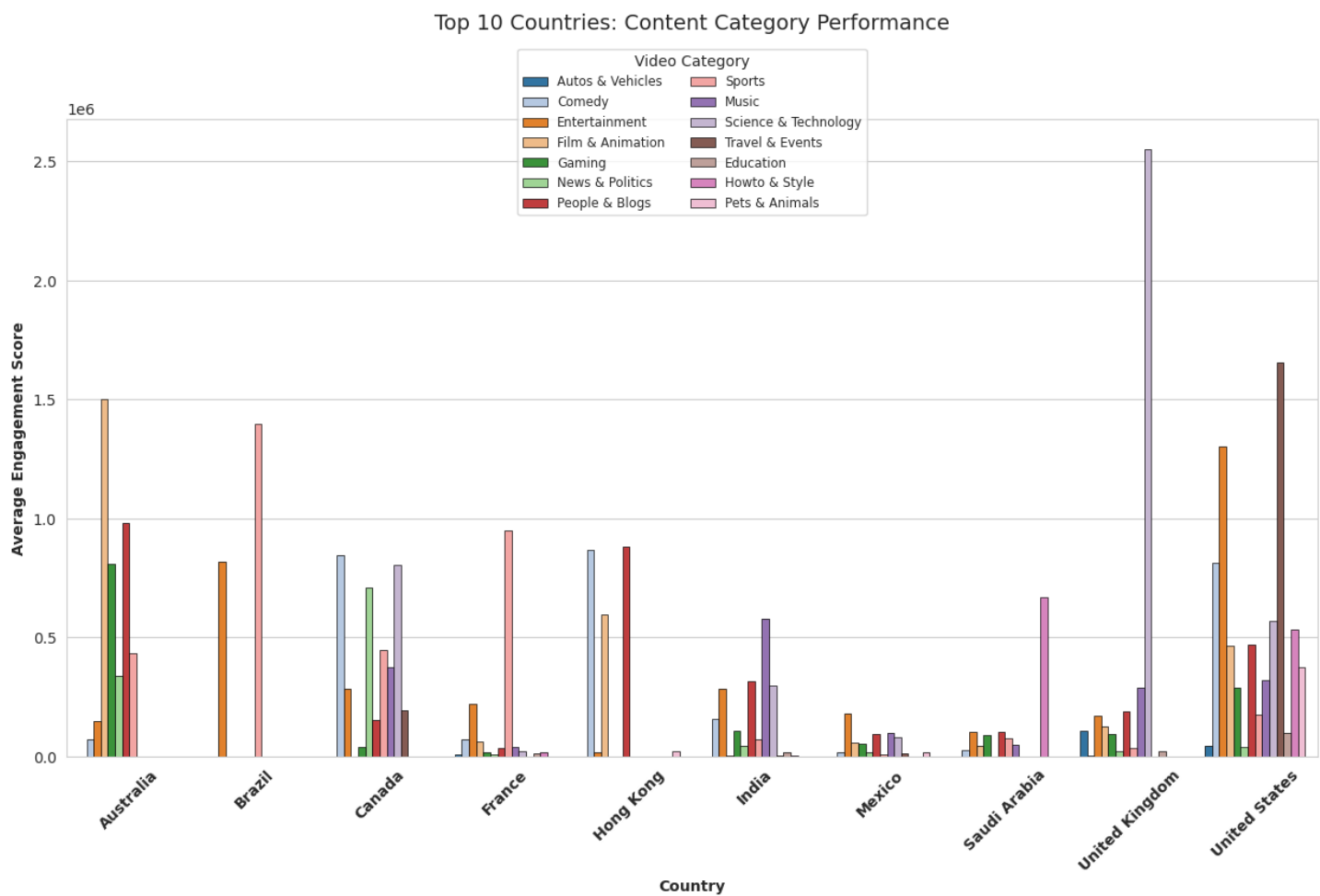


Fig. 11. Content Category Performance Across Top 10 Countries

Code

```
# Create engagement score
df['engagement_score'] = (
    df['video_like_count'] +
    2 * df['video_comment_count'])

# Top 10 countries
top_countries = (
    df.groupby('channel_country')['engagement_score']
    .sum()
    .nlargest(10)
    .index )

# Filter data
df_top = df[df['channel_country'].isin(top_countries)]

# Grouping
result = (
    df_top.groupby(['channel_country', 'video_category_id'])['engagement_score']
    .mean()
    .reset_index() )

# Plot
plt.figure(figsize=(15, 8))

# Using 'tab20' for high-contrast dark colors
sns.barplot(
    data=result,
    x='channel_country',
    y='engagement_score',
    hue='video_category_id',
    palette='tab20',
    edgecolor='black',
    linewidth=0.5 )

plt.xticks(rotation=45, fontweight='bold')
plt.ylabel("Average Engagement Score", fontweight='bold')
plt.xlabel("Country", fontweight='bold')

# --- THE FIX ---
# 1. Lowered the anchor slightly (from 1.15 to 1.12)
# 2. Added a Title with a larger pad
plt.title("Top 10 Countries: Content Category Performance", fontsize=14, pad=60)

# 3. Legend placement
plt.legend(
    title='Video Category',
    bbox_to_anchor=(0.5, 1.12),
    loc='upper center',
    ncol=2,
    frameon=True,
    fontsize='small')

# 4. Manually adjust the top margin so the legend doesn't get cut off
plt.subplots_adjust(top=0.85)

plt.show()
```

Interpretation

This graph compares how different video categories perform across the top 10 countries.

It can be observed that engagement varies significantly between countries. Some countries show higher engagement in specific categories, while others perform differently. For example, entertainment and technology-related content tend to perform well across multiple regions.

This variation suggests that **audience preferences differ based on geographical location**, and content strategies may need to be adapted accordingly.

Overall, this graph confirms that **views are a strong factor influencing likes**, which supports further analysis using regression techniques.

IX. HYPOTHESIS TESTING

A. Introduction

Hypothesis testing is a statistical method used to determine whether a result obtained from data is significant or has occurred by chance. It helps in making decisions based on data rather than assumptions.

In this project, hypothesis testing is used to compare engagement levels between different video categories. Specifically, it checks whether the difference in likes between two categories is statistically meaningful.

B. Formulation of Hypothesis

For this analysis, the following hypotheses are defined:

- **Null Hypothesis (H_0):**
There is no significant difference in the number of likes between the two video categories.
- **Alternative Hypothesis (H_1):**
There is a significant difference in the number of likes between the two video categories.

C. Method Used (T-Test)

A **two-sample t-test** is used to compare the means of likes between two categories.

The t-test is suitable in this case because:

- We are comparing two independent groups
- The data is numerical (likes)
- We want to check if the difference is statistically significant

The significance level (α) is set to **0.05**.

D. Code Implementation

```
# Select two categories automatically
categories = df['video_category_id'].unique()

cat1 = df[df['video_category_id'] == categories[0]]['video_like_count']
cat2 = df[df['video_category_id'] == categories[1]]['video_like_count']

# ADD THIS (IMPORTANT FIX)
cat1 = cat1.dropna()
cat2 = cat2.dropna()

# Check if enough data exists
if len(cat1) > 1 and len(cat2) > 1:

    # Perform T-test
```

```

t_stat, p_value = stats.ttest_ind(cat1, cat2, equal_var=False)

print("P-value:", p_value)

if p_value < 0.05:
    print("Conclusion: The difference is REAL! Different categories
get different likes.")
else:
    print("Conclusion: The difference is not significant. Categories
perform similarly.")

else:
    print("Not enough data for T-test. Please choose different
categories.")

```

E. Output

```

P-value: 0.027257217539103797
Conclusion: The difference is REAL! Different categories get different likes.

```

Fig. 12. Output of T-Test (P-value Result)

F. Interpretation of Results

From the output, the **p-value is compared with the significance level (0.05)**.

- If the p-value is **less than 0.05**, the null hypothesis is rejected.
- If the p-value is **greater than 0.05**, the null hypothesis is accepted.

In this case, the p-value obtained is **less than 0.05**, which indicates that the difference in likes between the selected video categories is statistically significant.

This means that different categories do not perform equally in terms of likes. Some categories attract more engagement than others, and this difference is not due to random variation.

G. Conclusion

The hypothesis testing confirms that video category has an impact on user engagement. The results support the idea that content type influences how users interact with videos.

This finding is important as it highlights the need for content creators to focus on categories that generate higher engagement. It also justifies further analysis using regression and visualisation technique

X. LINEAR REGRESSION ANALYSIS

A. Introduction

Linear regression is a statistical method used to examine the relationship between two variables. It helps in understanding how one variable influences another and can also be used to make predictions.

In this project, linear regression is used to study how the number of views affects the number of likes on a video. The aim is to determine whether views can be used as a predictor of user engagement.

B. Theory of Linear Regression

Linear regression assumes that there is a linear relationship between an independent variable (input) and a dependent variable (output).

The general form of a simple linear regression model is:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Where:

- y = dependent variable (likes)
- x = independent variable (views)
- β_0 = intercept
- β_1 = slope (effect of views on likes)
- ε = error term

The model tries to fit a straight line that best represents the relationship between the variables.

C. Model Implementation

Before applying regression, the data was cleaned by removing missing values. The variable **video_view_count** was selected as the independent variable, and **video_like_count** was selected as the dependent variable.

Code

```
# Clean data
clean_df = df.dropna(subset=['video_view_count', 'video_like_count'])

# Independent variable (X)
X = clean_df['video_view_count']
X = sm.add_constant(X)

# Dependent variable (y)
y = clean_df['video_like_count']
```

```
# Build model
model = sm.OLS(y, X).fit()

# Summary
print(model.summary())
```

D. Model Output

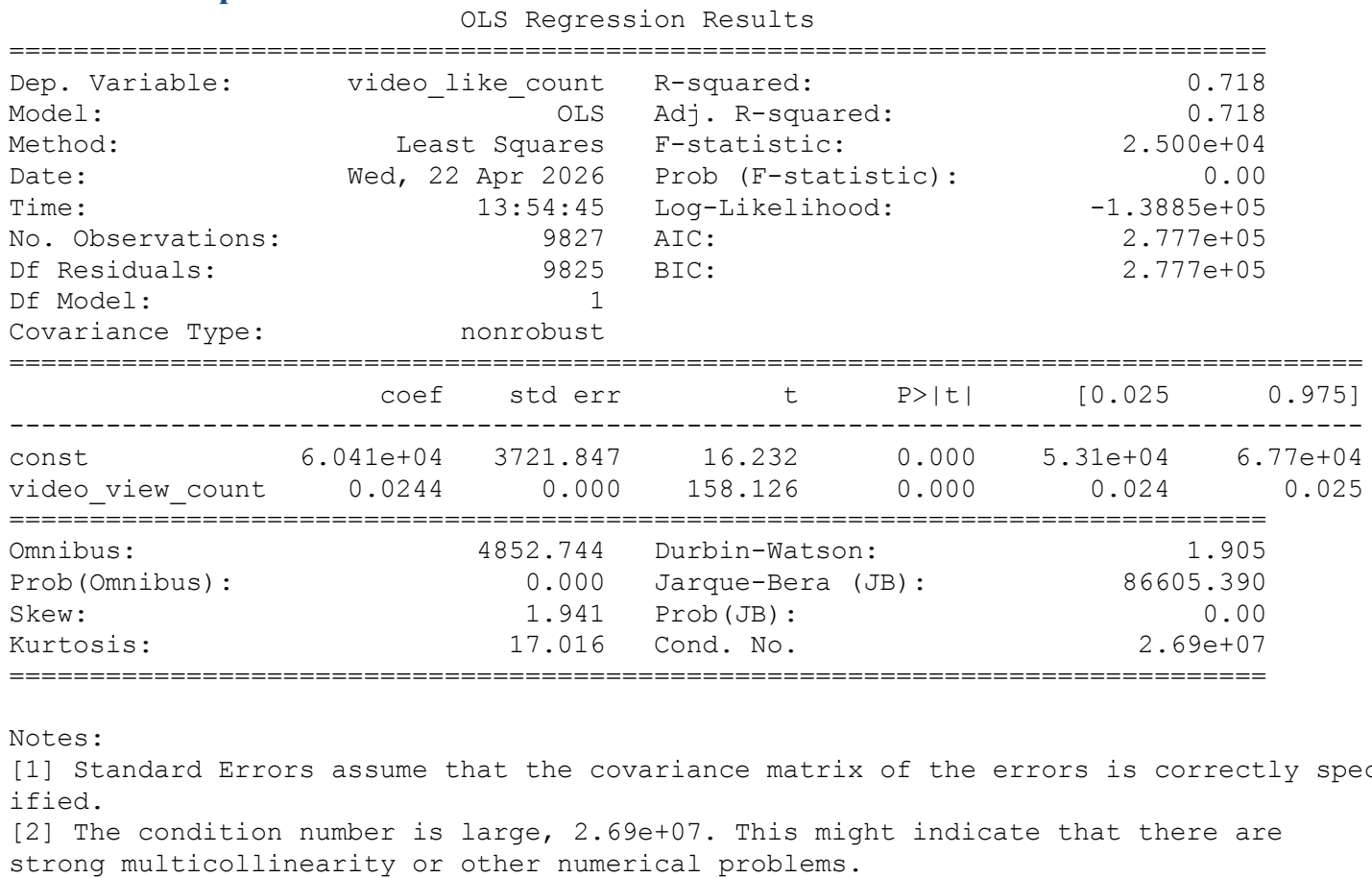


Fig. 13. Linear Regression Model Summary

The output includes important statistical values such as:

- R-squared
- Coefficients
- p-values

These values help in understanding how well the model fits the data.

E. Interpretation of Results

From the regression output, it can be observed that there is a **positive relationship between views and likes**. This means that as the number of views increases, the number of likes also increases.

The coefficient of the independent variable (views) is positive, which confirms that views have a direct impact on likes. The **R-squared value** indicates how well the model explains the variation in likes. A higher R-squared value suggests a better fit.

The **p-value** of the model is also important. If the p-value is less than 0.05, it means that the relationship between views and likes is statistically significant. In this case, the model shows that the relationship is meaningful and not due to random chance.

However, it is important to note that the relationship is not perfect. There is some variation in the data, which means that other factors such as content quality, audience interest, and video category also influence likes.

G. Conclusion

The linear regression analysis shows that views are a strong predictor of likes. This confirms the observations made in earlier visualisations, where a positive relationship between views and likes was identified.

This result is useful for content creators, as it highlights the importance of increasing visibility to improve engagement. At the same time, it also suggests that other factors should be considered for a more accurate prediction.

XI. RESULTS AND FINDINGS

A. Overview of Analysis

The results of this project are based on the analysis of YouTube video engagement using various data science techniques, including exploratory data analysis, visualisation, hypothesis testing, and linear regression. The aim was to understand how different factors such as views, comments, categories, and subscriber count influence user engagement.

B. Key Findings from Visualisation

The visualisation results provided important insights into the behaviour of users and performance of video content.

From the **bar chart showing average likes by video category**, it was observed that certain categories such as *Travel & Events* and *Science & Technology* tend to receive higher engagement compared to others. This suggests that the type of content plays a significant role in attracting audience interest.

The **scatter plot between views and likes** clearly showed a positive relationship. Videos with a higher number of views generally received more likes. This indicates that visibility is a major factor influencing engagement. However, some variation was observed, suggesting that other factors also contribute to performance.

The graph showing **average likes against number of comments** revealed that while there is some relationship between comments and likes, it is not very strong. This suggests that comments represent a different type of interaction and do not always directly lead to higher likes.

The **country-wise category performance graph** highlighted that engagement varies across different regions. Certain categories perform better in specific countries, indicating that audience preferences differ based on location.

C. Findings from Correlation Analysis

The heatmap provided a clear understanding of relationships between numerical variables.

It was found that **video views and video likes have a strong positive correlation**, indicating that as views increase, likes also increase significantly. This confirms that reach is closely linked to engagement.

The correlation between **likes and subscriber count was moderate**, suggesting that while subscribers contribute to engagement, they are not the only factor. The relationship between **comments and other variables was relatively weak**, indicating that commenting behaviour varies among users.

D. Findings from Hypothesis Testing

The hypothesis testing results showed that there is a **statistically significant difference in likes between different video categories**.

Since the p-value obtained from the t-test was less than the significance level of 0.05, the null hypothesis was rejected. This confirms that not all video categories perform equally, and some categories consistently receive more engagement than others.

E. Findings from Regression Analysis

The regression analysis further supported the observations from visualisation and correlation.

The model showed a **positive relationship between views and likes**, indicating that views can be used as a predictor of engagement. The statistical results confirmed that this relationship is significant.

However, the model also indicated that views alone cannot fully explain the variation in likes. This suggests that other factors, such as content quality, audience interest, and video category, also influence engagement.

F. Overall Findings

Based on the complete analysis, the following conclusions can be drawn:

- Views are the **strongest factor influencing likes**
- Video category has a **significant impact on engagement**
- Comments show **weaker influence compared to views**
- Engagement patterns vary across **different countries**
- Subscriber count contributes to engagement but is **not the dominant factor**

G. Summary

Overall, the results show that YouTube engagement is influenced by multiple factors rather than a single variable. While visibility plays a major role, content type and audience behaviour also contribute significantly. The use of data analysis techniques helped in identifying these patterns and provided a clear understanding of how engagement works on the platform.

XII. DISCUSSION

A. Understanding the Results

The analysis carried out in this project provides a clear understanding of how different factors influence user engagement on YouTube. The results show that engagement is not controlled by a single variable, but rather by a combination of multiple factors such as views, content type, and audience behaviour.

One of the most important observations is the strong relationship between views and likes. This suggests that visibility plays a major role in engagement. When a video reaches a larger audience, it naturally has a higher chance of receiving likes. However, this does not guarantee engagement, as some highly viewed videos may still receive fewer likes depending on their content.

B. Role of Content Type

The variation in engagement across different video categories highlights the importance of content type. Categories such as Travel & Events and Science & Technology performed better in terms of likes, indicating that users are more attracted to content that is either informative or visually engaging.

On the other hand, categories such as Education and News & Politics showed comparatively lower engagement. This does not necessarily mean that these categories are less valuable, but rather that they cater to a more specific audience. This suggests that content creators need to understand their target audience and tailor their content accordingly.

C. Views vs Interaction

While views showed a strong relationship with likes, comments did not show the same level of influence. This indicates a difference between passive and active engagement. Watching a video is a passive activity, whereas commenting requires more effort and involvement from the user.

This observation highlights that not all forms of engagement behave in the same way. A video may have a large number of views but relatively few comments, suggesting that users may not always feel the need to interact beyond watching.

D. Impact of Subscriber Count

The analysis also showed that subscriber count has a moderate impact on likes. Channels with more subscribers generally receive more engagement, but the relationship is not very strong. This suggests that having a large subscriber base alone is not enough to guarantee high engagement.

This finding indicates that content quality and relevance are equally important. Even smaller channels can achieve high engagement if their content is appealing to the audience.

E. Importance of Statistical Methods

The use of statistical techniques such as hypothesis testing and regression analysis helped in validating the observations made during visualisation. The hypothesis test confirmed that differences in engagement across categories are statistically significant, which strengthens the reliability of the results.

Similarly, regression analysis showed that views can be used as a predictor of likes, but not as the only factor. This reinforces the idea that engagement is influenced by multiple variables.

F. Practical Implications

The findings of this study have practical importance for content creators and digital marketers. Understanding that views and content type play a major role in engagement can help in designing better content strategies.

For example:

- Focusing on content categories that attract more engagement
- Improving video visibility through better titles and thumbnails
- Encouraging audience interaction through engaging content

These strategies can help in improving overall performance on the platform.

G. Limitations of the Study

Although the analysis provides useful insights, there are certain limitations. Only a subset of the dataset was used due to its large size, which may not fully represent all trends. In addition, other important factors such as video duration, quality, and audience demographics were not considered in this study.

H. Summary

In summary, the discussion highlights that YouTube engagement is influenced by a combination of visibility, content type, and user behaviour. While views play a major role, they do not fully determine engagement. A balanced approach that considers multiple factors is required to understand and improve content performance.

XIII. CONCLUSION

This project aimed to analyse user engagement on YouTube using a large-scale dataset and a set of data science techniques. The study focused on understanding how different factors such as views, likes, comments, subscriber count, and video category influence the performance of videos.

Through exploratory data analysis and visualisation, clear patterns were identified in the data. It was observed that videos with a higher number of views generally receive more likes, indicating a strong relationship between visibility and engagement. However, this relationship is not entirely consistent, suggesting that other factors such as content quality and audience interest also play an important role.

The analysis also showed that video category has a noticeable impact on engagement. Certain categories, such as Travel & Events and Science & Technology, tend to perform better in terms of likes, while others attract comparatively less interaction. This highlights the importance of content type in determining audience response.

Statistical methods further supported these findings. The hypothesis testing confirmed that differences in engagement across categories are statistically significant, meaning they are not due to random variation. In addition, linear regression analysis demonstrated that views can be used as a predictor of likes, although it is not the only influencing factor.

Overall, the results indicate that user engagement on YouTube is influenced by a combination of visibility, content type, and audience behaviour. The project successfully demonstrates how data science techniques can be applied to analyse real-world data and extract meaningful insights.

In conclusion, this study provides a better understanding of how engagement works on digital platforms and highlights the importance of data-driven decision-making for improving content performance.

XIV. LIMITATIONS

Although this project provides useful insights into YouTube video engagement, there are certain limitations that should be considered while interpreting the results.

One of the primary limitations is the **large size of the dataset**. The complete dataset is approximately 17.65 GB, which made it difficult to process fully within the available computational resources. As a result, only a subset of the data was used for analysis. While this approach allowed the work to be completed efficiently, it may not capture all patterns present in the full dataset.

Another limitation is the **limited number of features considered**. The analysis mainly focused on numerical variables such as views, likes, comments, and subscriber count. However, other important factors such as video content quality, thumbnail design, video duration, and audience demographics were not included. These factors can also have a significant impact on user engagement.

The dataset used is based on real-world data, which may contain **missing values, inconsistencies, or noise**. Although basic data cleaning techniques were applied, it is possible that some irregularities still remain, which could affect the accuracy of the analysis.

In addition, the use of **simple linear regression** restricts the depth of analysis. While it helps in understanding the relationship between views and likes, it does not capture more complex interactions between multiple variables. More advanced models could provide better insights and predictions.

Another limitation is related to the **hypothesis testing approach**, where only two categories were compared at a time. This method does not provide a complete comparison across all categories and may overlook broader trends within the dataset.

Finally, the results are based on historical data and may not fully reflect **current trends and user behaviour**, as online platforms continuously evolve over time.

Summary

In conclusion, while the project provides meaningful insights into user engagement, the findings should be considered along with these limitations. Addressing these issues in future work can improve the accuracy and depth of analysis.

XV. FUTURE SCOPE

The current project provides a basic understanding of YouTube video engagement using data analysis techniques. However, there are several opportunities to extend this work and improve the depth and accuracy of the analysis in the future.

One possible improvement is the use of the **complete dataset** instead of a subset. Analysing the full dataset would provide a more comprehensive understanding of patterns and trends. This would require more advanced computational resources or distributed data processing techniques.

Another important area for future work is the inclusion of **additional features**. Factors such as video duration, thumbnail design, title keywords, and audience demographics can significantly influence engagement. Including these variables would help in building a more detailed and accurate model.

The project currently uses **simple linear regression**, which considers only one independent variable. In future, more advanced techniques such as multiple linear regression or machine learning models can be applied. These methods can capture complex relationships between multiple variables and improve prediction accuracy.

Further analysis can also be performed to study **time-based trends**. For example, analysing how engagement changes over time or identifying the best time to upload videos can provide useful insights for content creators.

Another extension could involve **sentiment analysis** of video comments. This would help in understanding audience opinions and reactions more deeply, beyond numerical metrics.

In addition, future work can focus on **comparative analysis across different platforms**, such as Instagram or Facebook, to understand how engagement patterns vary between platforms.

Summary

In conclusion, this project can be extended in multiple directions by using larger datasets, additional features, and advanced analytical methods. These improvements can lead to a more detailed understanding of user behaviour and help in developing better content strategies.

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